TalentGuard AI

Preventing employee turnover with responsible AI

Hackathon Trusted AI March 2026

🔒 Cybersecurity

⚖️ Ethics AI

CONTEXT

The problem

33%

Turnover rate

of employees in the dataset have left

0

Predictive tools

HR teams lack systems to anticipate departures

HIGH

Legal exposure

HR data is sensitive — GDPR & AI Act apply

→ **Our answer: TalentGuard AI**

SOLUTION

How it works

01

GDPR Anonymization

02

Random Forest Model

03

NLP Sentiment Analysis

04

Fairness Audit AIF360

05

Simulator

Dashboard · Individual analysis · Fairness audit · Security · Risk simulator

Model

Random forest

	precision	recall	f1-score	support
Active	0.84	0.86	0.85	42
Left	0.70	0.67	0.68	21
accuracy			0.79	63
macro avg	0.77	0.76	0.76	63
weighted avg	0.79	0.79	0.79	63

Columns removed:
To avoid leaks:

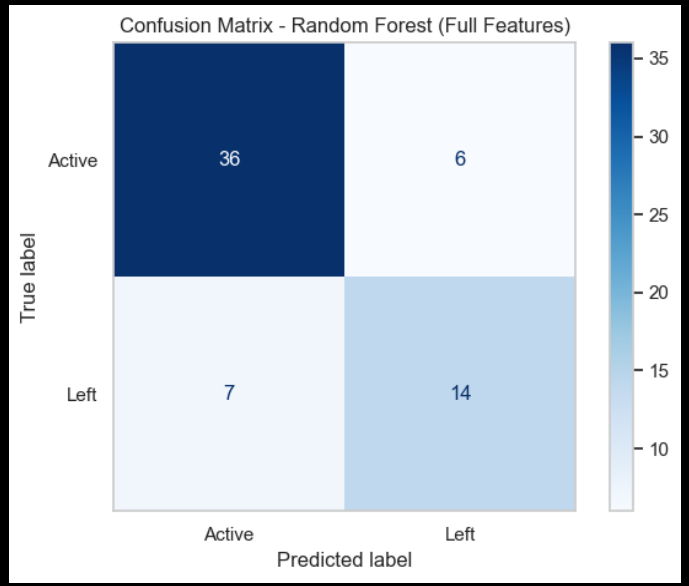
- DateofTermination
- EmpStatusID
- EmploymentStatus
- HasTermDate
- TermReason

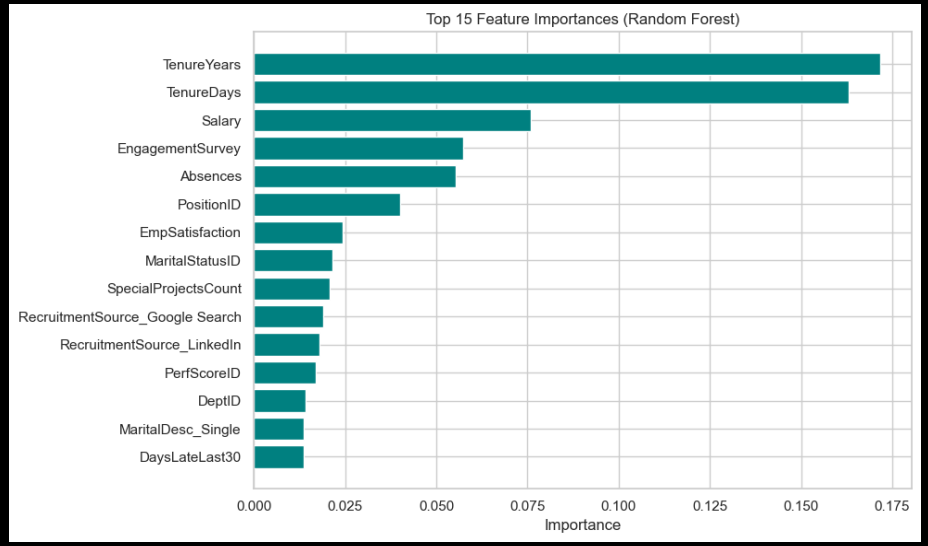
After creation of ternure:

- DateofHire
- LastPerformanceReview_Date

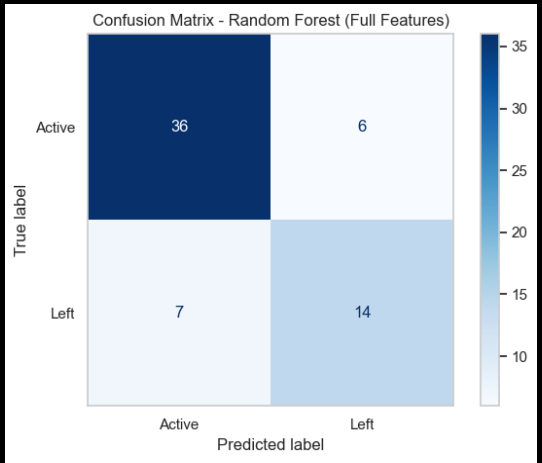
Becase they were not predictive:

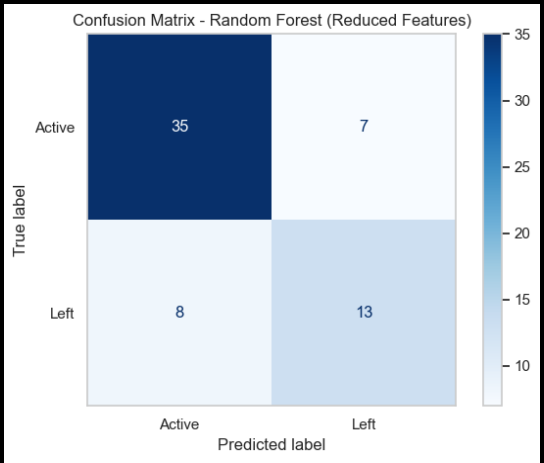
- ManagerID
- ManagerName





	precision	recall	f1-score	support
Active	0.81	0.83	0.82	42
Left	0.65	0.62	0.63	21
accuracy			0.76	63
macro avg	0.73	0.73	0.73	63
weighted avg	0.76	0.76	0.76	63





Employee Feedback Analysis with NLP and Machine Learning

Step 1 : 2 columns generated with AI to simulate employee comments (*ExitInterview_Feedback, InternalTransferRequest)*

Step 2 : lexicon-based sentiment analysis. Each comment is tokenized and scored : score = (positive_words –negative_words) / total_words

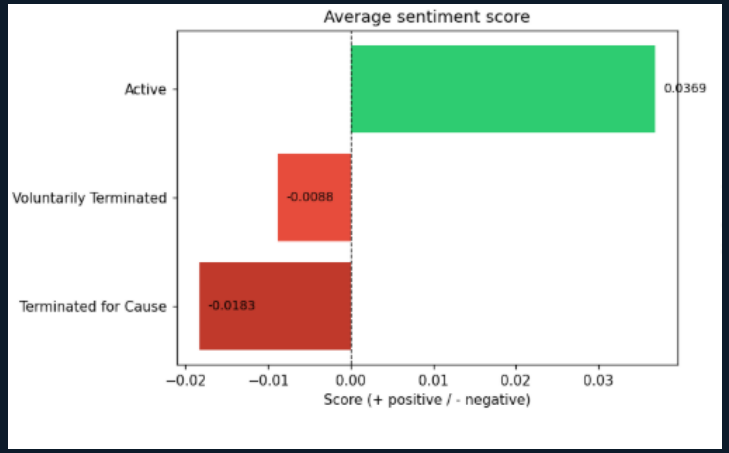
Step 3 : Text Vectorization (comments are transformed into numerical vectors with TF-IDF, weighting words by frequency and rarity.)

```python
POSITIVE_WORDS = {
    "positive", "appreciated", "valued", "satisfied", "happy", "proud",
    "collaborative", "supportive", "excellent", "strong", "growth",
    "recognition", "flexibility", "transparent", "engaged", "rewarding",
    "commended", "praised", "achievement", "alignment", "trust",
    "motivated", "welcoming", "enthusiastic", "warmly", "succeed",
    "opportunity", "benefited", "smooth", "genuine", "constructive",
    "effective", "open", "balance", "healthy",
}

NEGATIVE_WORDS = {
    "dissatisfied", "unhappy", "frustrated", "micromanaged", "overworked",
    "stagnant", "ignored", "unclear", "punitive", "conflict", "tense",
    "exhausted", "unresolved", "fear", "reprisal", "low", "broken",
    "negative", "concern", "complaint", "disappointment", "inadequate",
    "poorly", "misaligned", "leaving", "resigned", "resignation",
    "terminated", "departure", "lack", "unable", "untenable",
    "demotivating", "deterioration", "incompatible", "reluctantly",
    "disconnect", "opaque", "chronic",
}

NEGATION_WORDS = {"not", "no", "never", "without", "lack", "barely", "hardly"}

print(f"Positive lexicon: {len(POSITIVE_WORDS)} words")
print(f"Negative lexicon: {len(NEGATIVE_WORDS)} words")
```

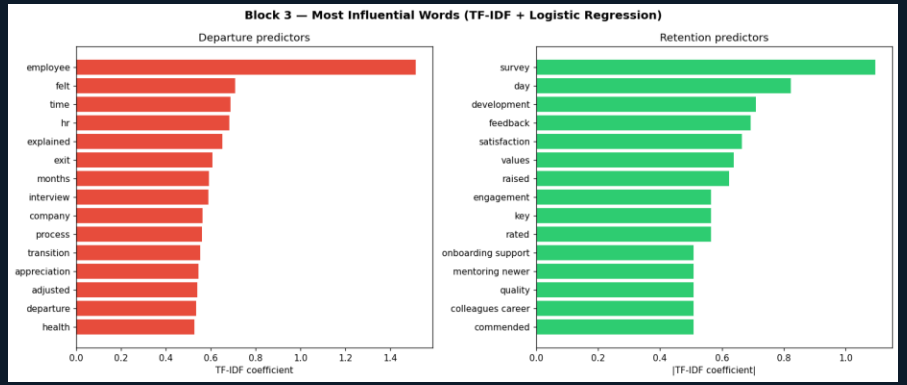


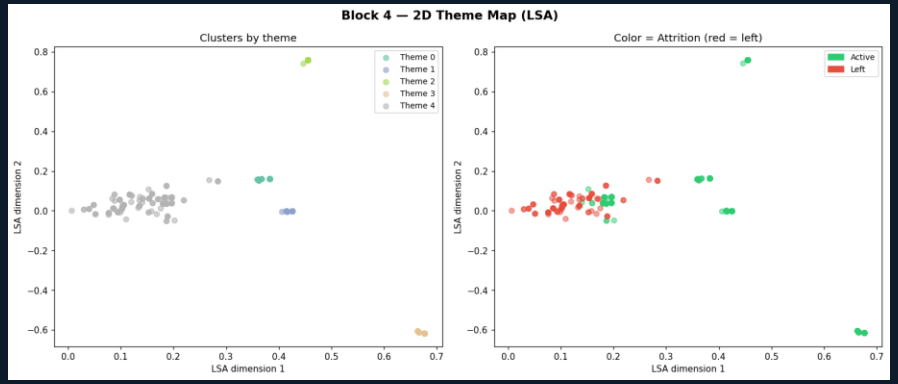
Employee Feedback Analysis with NLP and Machine Learning

Step 4 : Logistic Regression modeling (coefficients reveal words associated with leaving)

Step 5: Theme discovery with LSA and K-Means

Step 6: composite risk score





Protecting HR data

- ~311 employees (after anonymization)
- Target variable: Termd (0 = active, 1 = left)
- Turnover rate: ~33%
- Sensitive attributes present: Gender, RaceDesc, GenderID

Technique	Columns affected	Justification
Suppression	Employee_Name , ManagerName	Direct identifiers
Pseudonymization	EmpID	Truncated SHA-256 hash, non-reversible
Generalization	DOB → AgeGroup	Useful info without exact date
Masking	Zip → first 2 digits	Reduction of geographic precision
Model exclusion	Gender , RaceDesc , GenderID	Retained only for fairness audit

Fairness audit

Gender (M vs F)

-0.0183	+0.9748	-0.1213
Statistical Parity Diff	Disparate Impact	Equal Opportunity Diff
✅ OK	✅ OK	⚠ À surveiller
idéal : 0	idéal : 1	idéal : 0

Ethnicity (White vs others)

+0.0544	+1.0792	+0.1429
Statistical Parity Diff	Disparate Impact	Equal Opportunity Diff
✅ OK	✅ OK	⚠ À surveiller
idéal : 0	idéal : 1	idéal : 0

Shap

The application

Dashboard

Risk scores · RF vs NLP
Top 10 at-risk employees

Individual analysis

SHAP waterfall
Auto HR recommendation

Fairness audit

AIF360 metrics
Gender & ethnicity

Security

Input validation
Attack simulation

Risk simulator

5 sliders → live score
Frugal model

```
$ streamlit run app.py  →  http://localhost:8501
```